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Class :- SYIT	Batch :- B1
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Practical 25

Code -

```
object Jasad extends App {

  val sales: List[java.lang.Double] =
    List(100.0, 200.0, null, 300.0, null)

  val v = sales.filter(_ != null)

  val mean = v.map(_.toDouble).sum / v.length

  println("Mean Sales: " + mean)

  val n = sales.map { value =>
    if (value == null) mean
    else value
  }

  println("Original Data:")
  println(sales.mkString(", "))

  println("After Replacing Missing Values:")
  println(n.mkString(", "))

  println("Jasad S024")
}
```

Output -

```
"C:\Program Files\Java\jdk1.8.0_181\bin\java.exe" ...
Mean Sales: 200.0
Original Data:
100.0, 200.0, null, 300.0, null
After Replacing Missing Values:
100.0, 200.0, 200.0, 300.0, 200.0
Jasad S024
```

Practical 26

Code -

```
1  import breeze.linalg._
2
3  object Jasad extends App {
4      val x = DenseVector(1.0, 2.0, 3.0, 4.0, 5.0) // Dataset
5      val y = DenseVector(0.0, 0.0, 0.0, 1.0, 1.0) // Binary labels
6
7      // Initial weight and bias
8      var weight = 0.0
9      var bias = 0.0
10
11     // Sigmoid function
12     def sigmoid(z: Double): Double = {
13         1.0 / (1.0 + math.exp(-z))
14     }
15
16     // Train the model
17     for (i <- 1 to 1000) {
18         for (j <- 0 until x.length) {
19             val prediction = sigmoid(weight * x(j) + bias)
20             val error = prediction - y(j)
21
22             weight = weight - 0.01 * error * x(j)
23             bias = bias - 0.01 * error
24         }
25     }
```

```
26
27     // Classify the data
28     println("Predictions:")
29
30     for (i <- 0 until x.length) {
31         val probability = sigmoid(weight * x(i) + bias)
32
33         val result =
34             if (probability >= 0.5) 1 else 0
35
36         println("Input = " + x(i) +
37             ", Probability = " + probability +
38             ", Class = " + result)
39     }
40     println("Jasad S024")
41 }
```

Output -

```
"C:\Program Files\Java\jdk1.8.0_181\bin\java.exe" ...  
Predictions:  
Input = 1.0, Probability = 0.0515501057576029, Class = 0  
Input = 2.0, Probability = 0.16412826954644827, Class = 0  
Input = 3.0, Probability = 0.4149888799701166, Class = 0  
Input = 4.0, Probability = 0.7193154698480789, Class = 1  
Input = 5.0, Probability = 0.9025176083649726, Class = 1  
Jasad S024  
  
Process finished with exit code 0  
|
```